Métodos matemáticos para física. Curso 24/25.

Problemas ODE Tema 7. 10 puntos

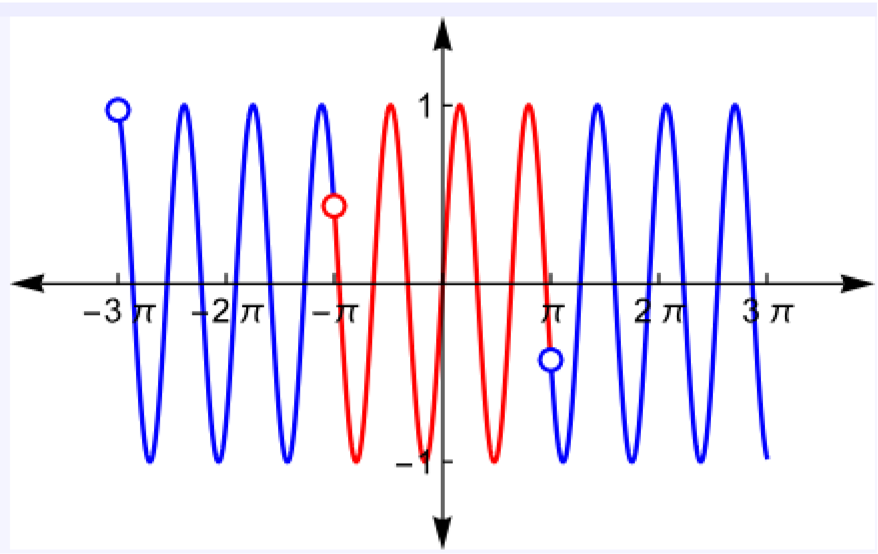
Problema 1 (4 puntos)

Supongamos que $f(t)$ se define en $(-\pi, \pi]$ como

a) $f(t) = \sin(\pi t)$.
b) $f(t) = \sin^2(t)$.
c) t^3
d) $f(t) = \begin{cases} -1 & \text{if } -\pi < t \leq 0, \\ 1 & \text{if } \quad 0 < t \leq \pi. \end{cases}$

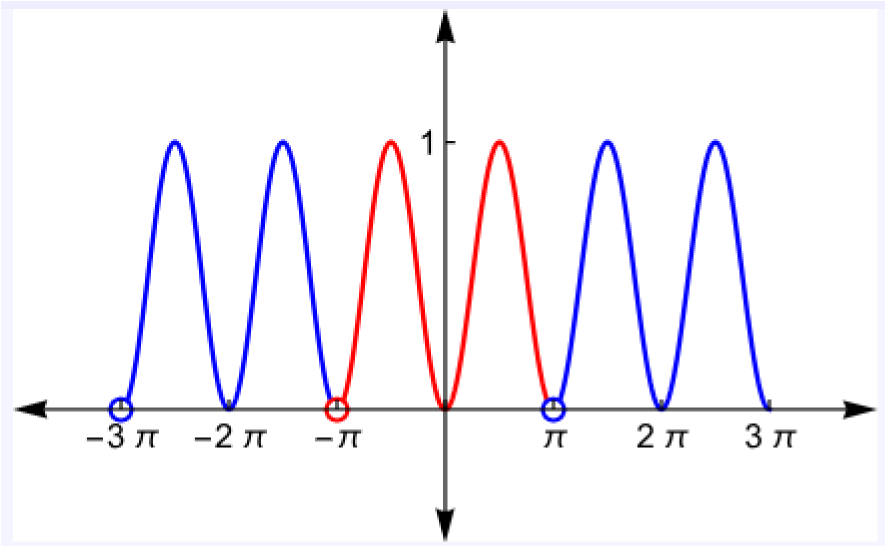
Extienda periódicamente y calcule la serie de Fourier.

a)



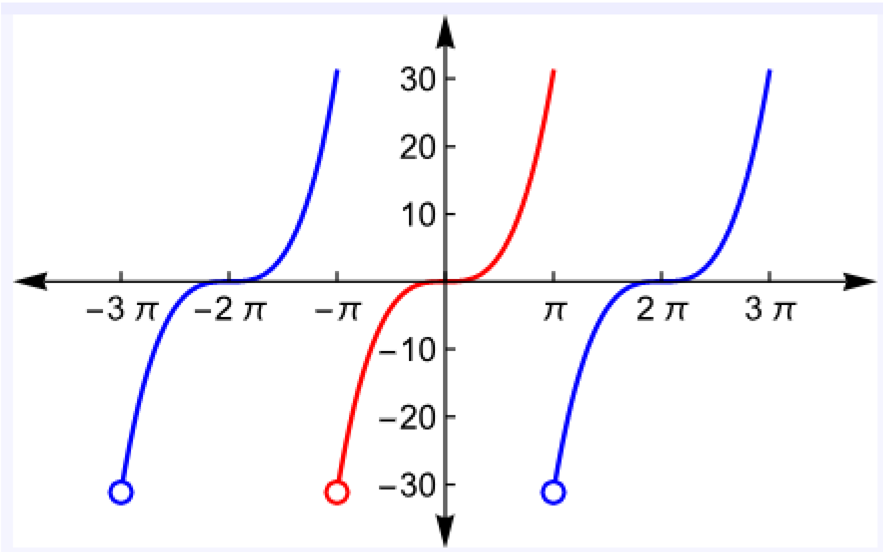
$$\sum_{n=1}^{\infty} \frac{(\pi-n)\sin(\pi n+\pi^2)+(\pi+n)\sin(\pi n-\pi^2)}{\pi n^2-\pi^3} \sin(nt)$$

b)



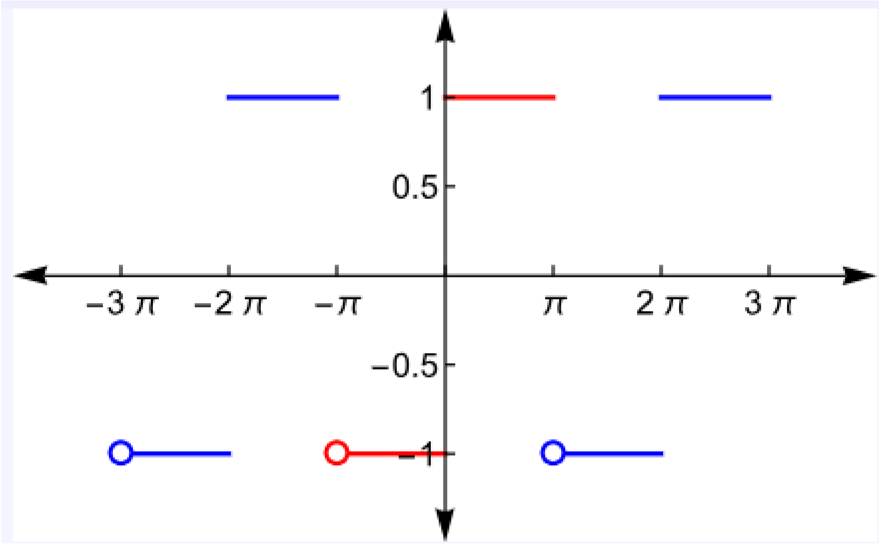
$\frac{1}{2} - \frac{1}{2}\cos(2t)$

c)



$\sum_{n=1}^{\infty} \frac{2(-1)^n(6-n^2\pi^2)}{n^3} \sin(nt)$

d)



$$\sum_{n=1}^{\infty} \frac{2(1-(-1)^n)}{n\pi}\sin(nt)$$

Problema 2 (2 puntos)

Sea $f(t) = t^2$ para $-2 < t \leq 2$ extendida periódicamente.

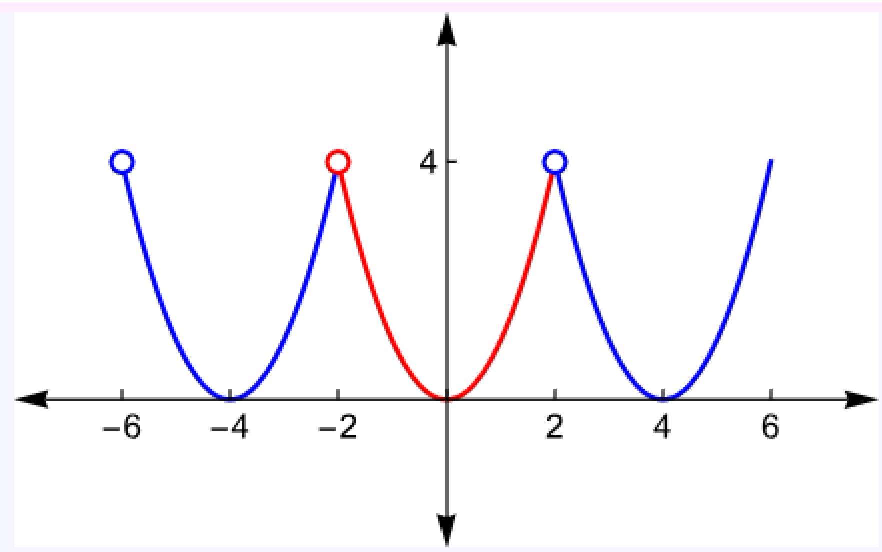
a) Calcule la serie de Fourier para $f(t)$.

b) Escriba la serie explícitamente hasta el tercer armónico.

a)

$$\frac{8}{6} + \sum_{n=1}^{\infty} \frac{16(-1)^n}{\pi^2 n^2} \cos\left(\frac{n\pi}{2} t\right)$$

$$\frac{8}{6} - \frac{16}{\pi^2} \cos\left(\frac{\pi}{2} t\right) + \frac{4}{\pi^2} \cos(\pi t) - \frac{16}{9\pi^2} \cos\left(\frac{3\pi}{2} t\right) + \cdots$$



Problema 3 (2 puntos)

Sea $f(t) = t \quad /3$ en el intervalo $0 \leq t < 3$.

a) Encuentra la serie de Fourier de la extensión periódica par.
b) Encuentra la serie de Fourier de la extensión periódica impar.

a)

$$\tfrac{1}{2} + \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{-4}{\pi^2 n^2} \cos\left(\frac{n\pi}{3} t\right)$$

b)

$$\sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{\pi n} \sin\left(\frac{n\pi}{3} t\right)$$

Problema 4 (2 puntos)

a) Dada la EDO $x''(t) + 2x(t) = f(t)$, donde $f(t) = t$ definida sobre el intervalo $0 < t < 1$. Dadas las condiciones de contorno $x(1) = 0$, $x(1) = 0$ encontrar la solución particular.

b) Repetir el problema para las condiciones de contorno $x'(0) = 0$, $x'(1) = 0$.

a)

$$f(t) = \sum_{n=1}^{\infty} c_n \sin(n\pi t).$$

$$c_n = 2\int_0^1 t\sin(n\pi t)\,dt = \frac{2\,(-1)^{n+1}}{n\pi}.$$

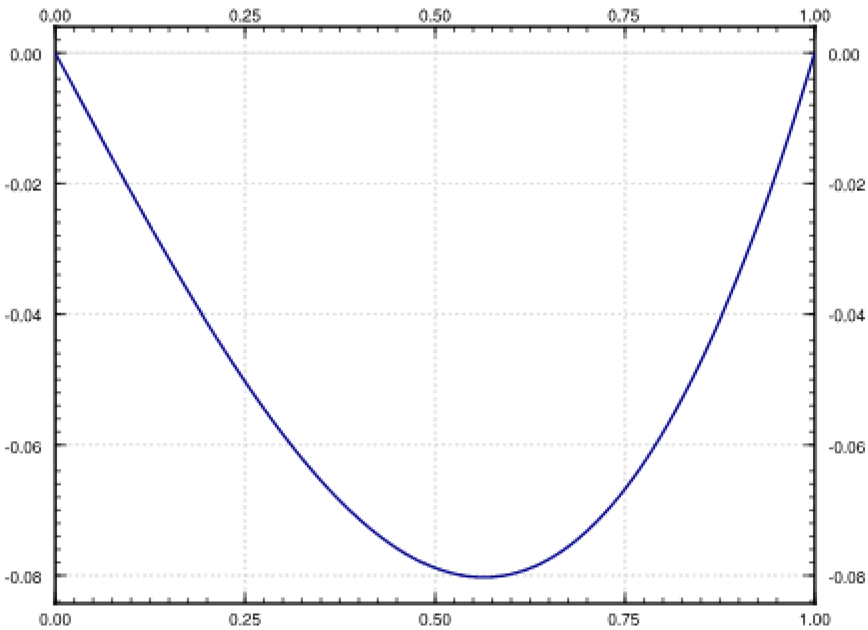
$$x(t) = \sum_{n=1}^{\infty} b_n \sin(n\pi t).$$

$$\begin{aligned} x''(t) + 2x(t) &= \underbrace{\sum_{n=1}^{\infty} -b_n n^2\pi^2 \sin(n\pi t)}_{x''} + \underbrace{2\sum_{n=1}^{\infty} b_n \sin(n\pi t)}_{x} \\ &= \sum_{n=1}^{\infty} b_n(2 - n^2\pi^2)\sin(n\pi t) \\ &= f(t) = \sum_{n=1}^{\infty} \frac{2\,(-1)^{n+1}}{n\pi} \sin(n\pi t). \end{aligned}$$

$$b_n(2 - n^2\pi^2) = \frac{2\,(-1)^{n+1}}{n\pi}$$

$$b_n = \frac{2\,(-1)^{n+1}}{n\pi(2 - n^2\pi^2)}.$$

$$x(t) = \sum_{n=1}^{\infty} \frac{2\,(-1)^{n+1}}{n\pi\,(2 - n^2\pi^2)} \sin(n\pi t).$$



b)

$$f(t) = \frac{c_0}{2} + \sum_{n=1}^{\infty} c_n \cos(n\pi t),$$

$$c_0 = 2\int_0^1 t\,dt = 1,$$

$$c_n = 2\int_0^1 t\cos(n\pi t)\,dt = \frac{2\big((-1)^n - 1\big)}{\pi^2 n^2} = \begin{cases} \frac{-4}{\pi^2 n^2} & \text{if } n \text{ odd,} \\ 0 & \text{if } n \text{ even.} \end{cases}$$

$$\begin{aligned} x''(t) + 2x(t) &= \sum_{n=1}^{\infty}\Big[-a_n n^2\pi^2\cos(n\pi t)\Big] + a_0 + 2\sum_{n=1}^{\infty}\Big[a_n\cos(n\pi t)\Big] \\ &= a_0 + \sum_{n=1}^{\infty} a_n(2 - n^2\pi^2)\cos(n\pi t) \\ &= f(t) = \frac{1}{2} + \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{-4}{\pi^2 n^2}\cos(n\pi t). \end{aligned}$$

Para valores pares

$a_0 = \frac{1}{2}, a_n = 0$

Para valores impares

$$a_n(2 - n^2\pi^2) = \frac{-4}{\pi^2 n^2},$$

$$a_n = \frac{-4}{n^2\pi^2(2 - n^2\pi^2)}.$$

$$x(t) = \frac{1}{4} + \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{-4}{n^2\pi^2(2 - n^2\pi^2)} \cos(n\pi t).$$